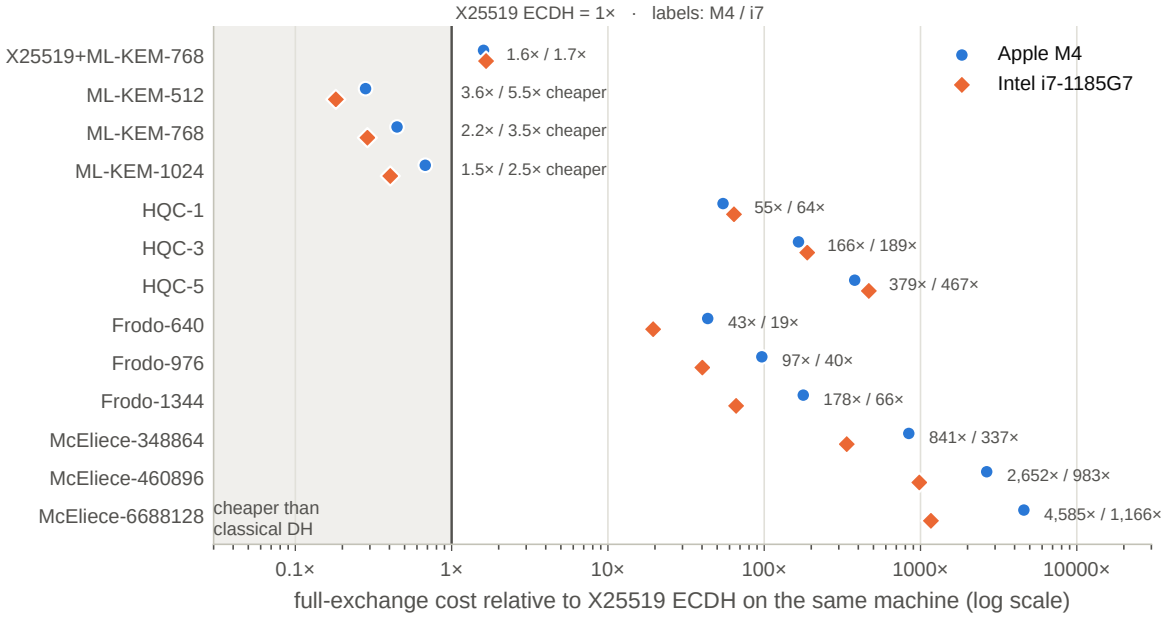
X25519 ECDH = 1× · labels: M4 / i7
Apple M4
Intel i7-1185G7
X25519+ML-KEM-768
ML-KEM-512
ML-KEM-768
ML-KEM-1024
HQC-1
HQC-3
HQC-5
Frodo-640
Frodo-976
Frodo-1344
McEliece-348864
McEliece-460896
McEliece-6688128
1.6× / 1.7×
3.6× / 5.5× cheaper
2.2× / 3.5× cheaper
1.5× / 2.5× cheaper
55× / 64×
166× / 189×
379× / 467×
43× / 19×
97× / 40×
178× / 66×
841× / 337×
2,652× / 983×
4,585× / 1,166×
cheaper than classical DH
0.1×
1×
10×
100×
1000×
10000×
full-exchange cost relative to X25519 ECDH on the same machine (log scale)